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Employee Attrition & Workforce Performance Analytics

Predictive Risk Analysis, Overtime Evaluation & Retention Strategies

Python (Pandas) Machine Learning Tableau Dashboard HR Metrics

1. Executive Project Overview

Analyzed 1,470 employee records to identify key drivers of workforce turnover. Applied Machine Learning models in Python to predict retention risks and proactively reduce hiring costs.

2. Key HR Metrics & Risk KPIs

TOTAL EMPLOYEES  
1,470  
Active Workforce

ATTRITION RATE  
16.12%  
237 Left

AVG INCOME  
\$6,503  
Benchmark

HIGHEST RISK  
Sales  
20.6% Attrition

3. Departmental Risk Breakdown

| Department             | Headcount | Attrition Count | Attrition Rate | Avg Overtime |
|------------------------|-----------|-----------------|----------------|--------------|
| Sales                  | 446       | 92              | 20.6%          | 18.4 hrs/mo  |
| Research & Development | 961       | 133             | 13.8%          | 12.1 hrs/mo  |
| Human Resources        | 63        | 12              | 19.0%          | 15.2 hrs/mo  |

4. Technical Implementation (Python ML)

```
import pandas as pd from sklearn.ensemble import RandomForestClassifier df = pd.read_csv('hr_employee_data.csv') X = df[['Age', 'MonthlyIncome', 'OverTime_Yes', 'YearsAtCompany', 'WorkLifeBalance']] y = df['Attrition_Yes'] model = RandomForestClassifier(n_estimators=100, random_state=42).fit(X, y) # Top Drivers: OverTime (32%), Monthly Income (28%), WorkLifeBalance (18%)
```

5. Strategic HR Recommendations

- 1. Address Overtime Burnout: Employees with regular overtime show 3x higher turnover (30.5% vs 10.4%). Restructuring saves ~\$180K annually.
- 2. Compensation Adjustment: Mid-level roles (Years 2-4) under \$3.5K salary drive 42% of churn. Targeted 8% revision retains top talent.